Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/27/2005
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Capillary Flow in Interior Corners		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Portland State University		
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Zip Code:	97207-0751	**Congressional District:**	3
Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/04/2002	**End Date:**	06/13/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rame, Enrique	**Contact Phone:**	216-433-2842
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Flight Program:			
Flight Assignment:	Note: PI changed organization/grant funded for flight		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2741		
Performance Goal No.:			
Performance Goal Text:			
	Capillary flows in interior corners have an established place in fluids-handling operations in reduced gravity environments. A quantitative understanding of corner flows is essential for the myriad fluid management tasks in space including flows in liquid-fuel tanks, heat pipes, energy systems, condensers, and life-support systems. The applications-oriented study is both experimental and theoretical to understand capillary-driven flows in containers with interior corners as well as flows in complex containers corner networks. For the experimental investigation, drop tower and low-g aircraft experiments are performed to confirm theoretical predictions of transient 3-dimensional fluid processes fundamental to many in-flight fuels/propellant management and thermal control problems. For the complex corner network experiments, flows in planar and subsequently three-dimensional geometries of increasing complexity will be quantified in terms of corner geometry, size, fluid volume, and fluid properties. The overall goal of the proposed		

Task Description:	research is to provide a firm experimental and theoretical footing for the design of systems exploiting dymnamic capillary phenomena in interior corners, corner networks, and complex containers. Current techniques rely on order of magnitude estimates and costly computational modeling. Capillary flows in interior corners occur routinely in low-g fluid systems in space. Many such flows, if not anticipated, can have ruinous consequences. A quantitative understanding of corner flows is essential for the myriad fluid management tasks in space including flows in liquid-fuel tanks, heat pipes, energy systems, condensers, life-support systems, and biofluids processes. A quantitative understanding of such flows is a valuable contribution to the aerospace community. Such a contribution will aid in innovation of new designs, shorten design cycles, enhance system performance, and increase system reliability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The first module of the Capillary Flow Experiment was launched to the International Space Station in January 2004 (P.I. Weislogel, Portland State University). The current performance date is expected to be around June and expected crew member Mike Fincke was trained at NASA JSC. The experiment investigates the extremes of the impact of the contact line condition for 'large' length scale capillary systems common aboard spacecraft. As part of a graduate course on Capillary Phenomena at Portland State University taught by M. Weislogel, a special project using K. Brakke's Surface Evolver algorithm was undertaken to compute the stability of wall-bound drops or bubbles that are also pinned at edges (so-called 'wall-bound edge-bound' drops and bubbles). During the project a new critical wetting condition was identified below which edge-bound drops are always stable. The critical contact angle is computed as a function of the gravity vector. The numerical procedure, which makes no account for contact angle hysteresis, predicts that such edge-bound drops are unconditionally unstable for any gravity field with a component that is tangent to the wall while inwardly normal to the edge. The numerical results are correlated simplifying the prediction of drop/bubble release conditions (often problematic) for a variety of terrestrial and microgravity fluids systems and will be submitted shortly for publication. Educational outreach arising from the NASA supported Capillary Flows research at Portland State U. continues to receive interest from the college, city, and state. P.I. M. Weislogel was selected as one of 3 featured scientists participating in this year's Intel International Science and Engineering Fair, hosted by Cleveland in 2003, but to be held in Portland in May 2004. During the program, Weislogel's work is highlighted which focuses on NASA related research and will air and be performed live to a target audience of 7000 Junior high students from local area schools. NASA GRC and MSFC are assisting the production company with materials and media including a GRC designed Mini-tower. Two invited presentations were delivered at the American Institute of Math Society's Pamona meeting on Dynamical Systems and Differential Equations by M. Weislogel and co-workeres from Portland State University;. A special session on Capillarity chaired by R. Finn (Stanford U.) included talks from noted speakers on large length scale capillary phenomena relevant to interfacial phenomena in low-g environments. The two talks in part presented by Weislogel focused on edge-bound and wall-bound drop and bubble stability, and a global similarity solution for capillary driven flows in complex polygonal containers. The water balloon rupture tests performed in NASA GRC's DC-9 appeared in the Gallery of Fluid Motion, Cambridge University Press. The book is a collection of the 'best of' flow visualization over the preceding decade. (M.M. Weislogel, S. Lichter, Water balloon rupture in low-g, Gallery of Fluid Motion, Cambridge University Press, eds: M. Samimy, P. Steen, K. Breuer, G. Leal, p 51, Jan. 2004.) Educational outreach arising from the NASA supported Capillary Flows research at Portland State University continues to receive interest from the college, city, and state. P.I. M. Weislogel was a featured scientist (one of three statewide) in this year's Intel International Science and Engineering Fair held in Portland in May 2004. During the program, Weislogel's work was highlighted which focused on NASA related research. The live program was performed before well over 6000 Junior and Senior high school students. DVD copies of the program are available.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Papers from Meeting Proceedings	Weislogel MM, Collicott SH, Gotti DJ, Bunnell CT, Kurta CE, Golliher E. "The Capillary Flow Experiments: Handheld Fluids Experiments for International Space Station." 42nd AIAA Aerospace Science Meeting and Exhibit, Reno, Nevada, January 5-8, 2004. 42nd AIAA Aerospace Science Meeting and Exhibit, Reno, Nevada, January 5-8, 2004. Paper AIAA 2004-1148. https://doi.org/10.2514/6.2004-1148 , Jan-2004
Papers from Meeting Proceedings	Weislogel MM, Nardin CL. "Passive Fluids Management in Low-g: Partially Wetting Systems." 42nd AIAA Aerospace Sci. Meeting and Exhibit, Reno, Nevada, January 4-8, 2004. 42nd AIAA Aerospace Sci. Meeting and Exhibit, Reno, Nevada, January 4-8, 2004. AIAA paper 2004-1152. https://doi.org/10.2514/6.2004-1152 , Jan-2004
Presentation	1. M.M. Weislogel "A global similarity solution for spontaneous capillary flow in cylinders of right polygonal section" AIMS' Fifth International Conference, Dynamical Systems and Differential Equations, Cal Poly, Pamona, CA June 16-19, 2004 Jun-2004
Presentation	2. Y.K. Chen, M. Bacich, C. Nardin, A. Sitorus, M.M. Weislogel "The shape and stability of wall-bound drops and bubbles pinne3d along straight edges" AIMS' Fifth International Conference, Dynamical Systems and Differential Equations, Cal Poly, Pamona, CA June 16-19, 2004 Jun-2004
Significant Media Coverage	Weislogel MM. "PSU sends experiment to space. Good Day Oregon Program." Good Day Oregon Program. KPTV Channel 12, Portland, 5:05pm, 6:08pm, and 8:05pm, January, 2004., Jan-2004